

Anubhav

Codec numbers + the story

GTX 1650

Stage 2 PoC - 1-2 Sept

SIH26104 - Team Sonix

Your mission: Prove the codec claim with numbers, and own how we tell the story on stage.

THE SITUATION - READ THIS FIRST

Where we are. SONIX listens to a call in 4-second chunks and shows green / amber / red. On the academic benchmark it is strong: **1.49% EER** on ASVspoof-2019 LA. But on our own real recordings it wrongly flags roughly **half of genuine voices as fake**.

Why. The model only ever learned from clean studio recordings of real people - same good microphone, same quiet room. Our phone and laptop recordings sound nothing like that. So the model thinks: "this is not like any real voice I know, therefore fake." **The model is not broken. It is narrow.** It has never heard normal life.

The fix - three things. (1) **Show it more kinds of real:** add genuine recordings from many phones, mics, rooms and languages to training. (2) **Make it ignore silence:** quiet gaps are what break the "minimal voice" clips. (3) **Reset the alarm levels:** our amber/red numbers came from the clean test, so we re-set them using real recordings.

HARD RULE: we never train on In-the-Wild or on any eval set. Those are our honest test. Training on them would destroy the one number that proves we generalise.

THE PLAN - THREE LANES RUNNING AT ONCE

LANE 1 - FIX THE MODEL

Silence gate, more real audio, RawBoost augmentation, retrain, recalibrate. This is what stops the false alarms.

Yugal - Akshat - Yukti

LANE 2 - HONEST NUMBERS

2021-DF EER, In-the-Wild EER, codec 2x2, metrics + calibration. These are the numbers we quote to the jury.

Navya - Anubhav - Yukti

LANE 3 - A DEMO THAT CANNOT FAIL

Replay mode on validated clips, clip library, pitch, Q&A, backup video. The demo never depends on live luck.

Suryansh - Anubhav - Akshat

YOUR TASKS

P1 TONIGHT

Codec the eval set and extract

- Produce G.711 versions of the eval audio, then extract their embeddings.
- Your 1650 is slower - start this first and let it run while you write the pitch.

```
python make_codec.py --split eval
python src\extract_embeddings.py --split eval --data-root data\asvspoof19_la_g711 -
-out outputs\embeddings_g711 --batch 8
```

Done when: G.711 eval embeddings exist.

P1
TOMORROW

Score the codec 2x2

- Four numbers: {baseline, augmented} x {clean eval, G.711 eval}.
- The story we want: the augmented head degrades far less when audio is phone-compressed.

```
python src\eval.py --split eval --model-ckpt outputs\models\head.pt
python src\eval.py --split eval --model-ckpt outputs\models\head_aug.pt
# repeat both with eval pointed at the G.711 embeddings
```

Done when: A filled 2x2 table of EERs.

P1 TONIGHT

Opening pitch, 2-3 minutes

- Caller ID is fakeable. The voice is what the attack uses. Fraud systems only notice afterwards. The victim gets no signal.
- SONIX gives that signal while the call is live.
- Then: how it works in one line -> 1.49% in-domain -> honest about real-world limits -> it never blocks a call, a human always decides.

Done when: A written script you can deliver without notes.

P2
TOMORROW

Jury Q&A prep

- Write 10 likely questions with honest answers. The three that WILL come up:
- 1) 'Why not AASIST like your PPT says?' -> our methodology slide stages it: baseline first, then augmentation, then one-class. AASIST needs frame-level features, a pipeline rebuild, and it is our next stage.
- 2) 'Does it work on real calls?' -> honestly: in-domain 1.49%, real-world is the field's open problem, which is why we never block and always flag to a human.
- 3) 'Is 1.49% just accuracy?' -> no, it is EER on a public benchmark, externally comparable.

Done when: 10 Q&A pairs written down.

YOUR CHECKLIST

- ☐ G.711 eval embeddings
- ☐ Codec 2x2 table filled
- ☐ Pitch script written
- ☐ 10 Q&A pairs ready

Note: Confirm the G.711 eval data-root name with Yugal before extracting.

WORDS WE KEEP USING

EER	One number for how wrong the model is. Lower = better. 1.49% is strong. 50% = coin flip.
bonafide	A real human voice.
spoof	An AI-cloned / synthetic voice.
embedding	The 1024-number fingerprint of a 4-second audio chunk.
head	The small model we actually train. The big wav2vec2 front-end is frozen (never trained).
VAD	Voice Activity Detection - is anyone actually speaking in this chunk?

augmentation	Making altered copies of training audio (phone-compressed, noisy) so the model sees variety.
calibration	Making the 0-1 score mean a real probability, so thresholds are meaningful.
domain shift	Test audio looks different from training audio. THIS is our core problem.